Do Now · retrieval

Autumn mock feedback and intervention

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why must binary search use sorted input?

Q2

What is a base case?

Q3

Why do exam questions award method marks?

Q4

A rising-edge D flip-flop samples D=1 at the first clock edge. D then changes to 0 between edges, and another rising edge occurs. State Q immediately after the first edge, between edges, and after the second edge.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Diagnose the weakness — Look for a pattern across the marked questions: omitted intermediate states, incorrect notation or missed explanations.

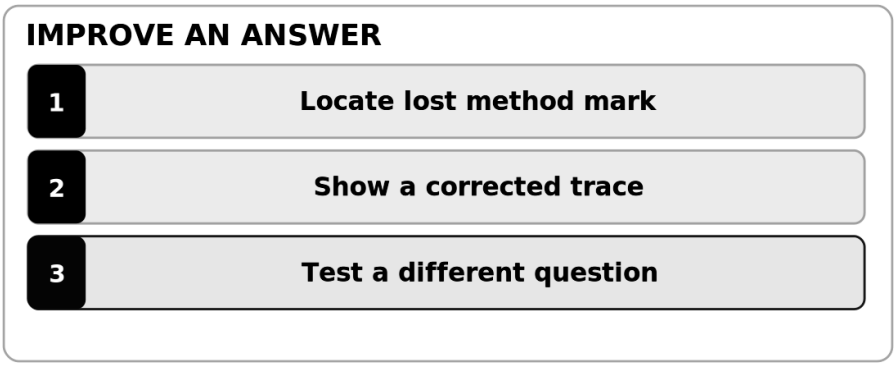
Core idea 2

Rebuild the method — Model each calculation, comparison or state change instead of simply copying the final answer.

Core idea 3

Test transfer — Give students a different array or bit pattern and check whether they can produce the same method independently.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

diagnostic — Assessment used to locate a learning gap.

method mark — Credit for a correct intermediate step.

transfer — Applying a method to a changed problem.

correction — Justified replacement for incorrect reasoning.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

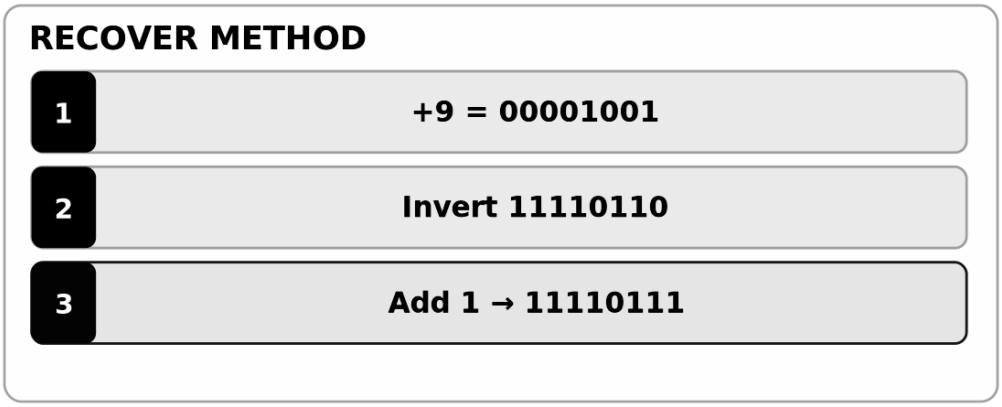
Correct an incomplete two's-complement answer for −9 in eight bits.

Follow the worked method

1. Predict first — Start with +9 in exactly eight bits and invert it.

2. Explain the answer — Add one to give 11110111, representing −9. This explicitly shows the stages.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Compare both candidate routes, recording their total costs.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Improve an answer to 'trace linear search for 7 in [3,5,7,9]' by showing comparisons and returned index.

Question 2

Correct the claim '−6 is 11111001 in eight-bit two's complement'. Provide the full method.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A student traces binary search for 13 in [1,3,5,7,9,11,13] but does not change high or low after a comparison. Rewrite the trace with bounds and explain why it terminates.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Copying only a mark-scheme answer proves you can repeat the method.

Use key vocabulary from memory

1. Explain diagnostic in your own words.

2. Explain method mark in your own words.

3. Explain transfer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.